

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

AM2221-1 – EMPOWERING WITH ARTIFICIAL INTELLIGENCE

(For AM)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What is the primary focus of the AI ethics issue "privacy"?**
 A) Fair treatment of AI systems B) Ensuring data security and confidentiality
 C) Environmental impacts of AI D) Providing access to AI tools for all
- Which of the following tasks is part of Data Acquisition in an AI project?**
 A) Identifying stakeholders B) Collecting and importing data
 C) Creating a neural network D) Predicting outcomes
- What does the term "unsustainable development" in AI refer to?**
 A) AI systems that cannot evolve B) AI systems without ethics
 C) AI systems with negative environmental or social impacts D) AI systems that require constant human intervention
- What is the role of neural networks in AI?**
 A) To process and predict complex patterns in data B) To classify ethical AI models
 C) To create privacy filters in AI D) To store large amounts of data
- Which machine learning technique is commonly used for data prediction tasks?**
 A) Support Vector Machine (SVM) B) K-Nearest Neighbors (KNN)
 C) Decision Trees D) All of the above
- What is a key aspect of AI in industries?**
 A) Solving simple tasks manually B) Automating complex processes and decision making
 C) Ensuring bias in decisions D) Limiting data collection
- In computer vision, how is an image represented in numbers?**
 A) As pixels based on intensity values B) As text files
 C) Using binary search D) As decision trees
- Which of the following is a basic technique in computer vision?**
 A) Tokenization B) Thresholding
 C) Bag of Words D) Sentiment Analysis
- What does "region of interest" (ROI) mean in computer vision?**
 A) The most important part of an image for processing B) The Entire Image
 C) The boundary pixels of an image D) The histogram of the image
- Which of the following is NOT a geometric transformation used in image processing?**
 A) Resizing B) Cropping
 C) Tokenization D) Rotating

11. The K-Nearest Neighbors (KNN) algorithm in computer vision is primarily used for:
 - A) Clustering
 - B) Classification
 - C) Data pre-processing
 - D) Regression
12. Which algorithm is commonly used for separating hyperplanes in classification tasks?
 - A) Support Vector Machine
 - B) K-Nearest Neighbor (KNN)
 - C) Neural Networks
 - D) Decision Trees
13. How does OpenVINO assist in computer vision tasks?
 - A) By collecting training data
 - B) By optimizing inference performance of pre-trained models
 - C) By training models from scratch
 - D) By enhancing the resolution of images
14. What is the purpose of "data pipelining" in NLP?
 - A) To store data in a database
 - B) To visualize language models
 - C) To create feature vectors
 - D) To automate the process of data preprocessing and model execution
15. What is the function of the TF-IDF method in NLP?
 - A) To assign importance to words based on their frequency in a document relative to the corpus
 - B) To find the most frequent words in a document
 - C) To generate sentences from a chatbot
 - D) To create deep learning models
16. Which Python library is commonly used for machine learning models in NLP?
 - A) NumPy
 - B) TensorFlow
 - C) Scikit-learn
 - D) OpenCV
17. A chatbot's specialty refers to:
 - A) Its ability to change tasks
 - B) The specific domain or area it is designed to interact with
 - C) The number of users it can handle
 - D) Its sentiment analysis capabilities
18. What is the main advantage of the Bag of Words model in NLP?
 - A) It is easy to implement and provides basic feature extraction
 - B) It captures the word order in the document
 - C) It improves the accuracy of neural networks
 - D) It create embeddings for all words
19. Which technique in NLP converts words into numerical representations?
 - A) Geometric transformation
 - B) Word embeddings
 - C) Feature extraction
 - D) Image preprocessing
20. Which of the following is a common method of visualizing statistical data?
 - A) KNN Algorithm
 - B) Neural Networks
 - C) Chatbots
 - D) Histograms

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the phases of AI project Life Cycle.	8	L*2	1	1
	b) Explain how to handle erroneous data and missing data in machine learning using a code snippet.	8	L2	2	1
2.	a) Discuss the three classifications of Machine Learning.	8	L2	1	1
	b) Explain the Terms:				
	(i) Overfitting	4	L2	1	1
	(ii) Underfitting	4	L2	2	1
	c) Explain the processing layers in Artificial Neural Networks.	4	L2	2	1
3.	a) List and explain the goals of Sustainable Development Goals.	8	L2	1	1

- b) Calculate the Entropy and Information Gain for the given set

Weather Condition	Play Tennis(YES)	Play Tennis(NO)	Total
Sunny	2	3	5
Overcast	4	0	4
Rainy	3	2	5
Total	9	5	14

8 L3 2 2

Unit – II

4. a) What is a Contour? Write a code snippet to detect the contours in the image. 8 L3 3 2
 b) Discuss the K – Nearest Neighbor (KNN) Algorithm. 8 L2 4 1
5. a) Explain Support Vector Machine Terminologies and how does that work. 8 L2 3 1
 b) Create a canvas with 2 Boxes. The first box says "ACCESS GRANTED" and the second box says "PLEASE PROCEED" 8 L3 4 2
6. a) Explain how a computer processes and sees an image. Discuss the role of the RGB color model in image representation. 8 L2 3 2
 b) Describe the importance of geometric transformations such as resizing, cropping and rotating in computer vision tasks. How do these transformations affect the analysis of images? 8 L1 4 1

Unit – III

7. a) What are the steps of Text Normalization? Explain them in Brief. 8 L2 5 1
 b) Normalize the given text and comment on the vocabulary before and after the normalization:
 Raj and Vijay are best friends. They play together with other friends. Raj likes to play football but Vijay prefers to play online games. Raj wants to be the footballer. Vijay wants to become an online gamer. 8 L3 5 2
8. a) Through a step-by-step process, calculate TFIDF for the given corpus and mention the words having highest value.
 A. Document 1: We are going to Mumbai
 B. Document 2: Mumbai is a famous place
 C. Document 3: We are going to a famous place
 D. Document 4: I am famous in Mumbai. 8 L3 5 2
- b) Write a short note on:
 (i) Sentiment Analysis
 (ii) Inverse Document Frequency
 (iii) Hash map
 (iv) Bag of Words 8 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
